

Zhenyuan Wang

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PROFESSIONAL EXPERIENCE

Infrared Processing and Analysis Center (IPAC), Caltech

Postdoctoral Scholar Research Associate

Supervisor: Dr. Yun Wang

Pasadena, CA, US

Sep. 2025 – now

The Pennsylvania State University

Graduate Research Assistant

Supervisor: Prof. Donghui Jeong

State College, PA, US

Aug. 2019 – Aug. 2025

EDUCATION

The Pennsylvania State University

Ph.D. in Astronomy and Astrophysics

Thesis: *Forward Modeling Large-scale Structure in Galaxy Surveys with Perturbation Theory*

Thesis Advisor: Prof. Donghui Jeong

State College, PA, US

Aug. 2019 – Aug. 2025

The Pennsylvania State University

Pre-Doctoral Student

Advisor: Prof. Donghui Jeong

State College, PA, US

Aug. 2018 – Dec. 2018

MIT, Kavli Institute for Astrophysics and Space Research

Research Internship

Advisor: Prof. Mark Vogelsberger & Dr. Rahul Kannan

Cambridge, MA, US

June 2017 – Sept. 2017

Tsinghua University

B.S. in Physics

Advisor: Prof. Yi Mao, Prof. Houjun Mo

Beijing, China

Aug. 2014 – July 2018

RESEARCH INTEREST

Large-scale structure, cosmological perturbation theory, perturbative galaxy bias expansion, redshift-space distortion, SNe Ia cosmology, data denoising.

PUBLICATION LIST

- [1] *Modeling Biased Tracer Field with Renormalized Galaxy Bias Expansion*
Wang, Z., Jeong, D., Taruya, A., Nishimichi, T., & Osato, K. (2026).
In preparation
- [2] *Lossless Compression of Cosmological Information from Type Ia Supernova Distance Measurements*
Wang, Z., & Wang, Y. (2026).
Submitted to JCAP, arxiv:[\[2605.19188\]](#)
- [3] *Model-Independent Analysis of Type Ia Supernova Datasets and Implications for Dark Energy*
Wang, Z., & Wang, Y. (2026).
Submitted to JCAP, arxiv:[\[2604.11883\]](#)
- [4] *Perturbation theory remixed II: Improved modeling of nonlinear matter bispectrum*
Wang, Z., Jeong, D., Taruya, A., Nishimichi, T., & Osato, K. (2024).
[Physical Review D](#), 110, 103548, arxiv:[\[2408.06413\]](#)
- [5] *Recovering the Missing Large-Scale Density Modes in 21cm Intensity Mapping from the Scalar-Type Clustering Fossils*
Wang, Z., & Jeong, D. (2024).
[JCAP](#), 07(2024)020, arxiv:[\[2312.17321\]](#)

- [6] *Perturbation theory remixed: Improved nonlinearity modeling beyond standard perturbation theory*
Wang, Z., Jeong, D., Taruya, A., Nishimichi, T., & Osato, K. (2023).
Physical Review D, 107(10), 103534, arxiv:[2209.00033]
- [7] *The Breakdown Scale of HI Bias Linearity*
Wang, Z., Chen, Y., Mao, Y., Mo, H., Wang, H., Guo, H., ... & Zheng, Z. (2021).
The Astrophysical Journal, 907, 4, arxiv:[1901.02724]

AWARDS & ACHIEVEMENTS

Zaccheus Daniel Fellowship (2024)
Downsbrough Graduate Fellowship in Astrophysics for outstanding research (2024)
First Prize in KIAS QUG Summer School on A.I. in High Energy Physics (2023)
 Project: *Train Generative Adversarial Network to generate LHC events*
 Mentor: **Dr. Ramon Winterhalder**
Zaccheus Daniel Fellowship (2022)
Homer F. Braddock / Nellie H. and Oscar L. Roberts Fellowship (2019)
National Astronomical Observatories, Chinese Academy of Sciences Fellowship (2017)

CONFERENCE PRESENTATIONS & TALKS

Talk: Robust Modeling of Galaxy Large-scale Structure (30 min)
 Institute for Advanced Study/Princeton University Early Universe/Cosmology Lunch Discussion (2024)

Talk: Towards Accurate Galaxy Bias Renormalization (30 min)
 YITP International molecule-type workshop: **Theory and Data Analysis Challenges for Cosmological Large-Scale Structure Observations** (2024)

Seminar: Perturbative Modeling of Galaxy Large-scale Structure (40 min)
 USM Galaxies and Lensing Seminar, Ludwig Maximilian University of Munich (2024)

Talk: Perturbation Theory Remixed
 BCCP cosmology workshop: Cosmology in the Adriatic – From PT to AI (2024) [Slides Link]

Talk: Perturbation Theory Remixed
 Workshop: New Strategies for Extracting Cosmology from Galaxy Surveys - 2nd edition (2024) [Slides Link]

Seminar: Perturbative Modeling of Galaxy Large-scale Structure (40 min)
 Department of Astronomy, Shanghai Jiaotong University (2024)

Talk: The Renormalization of Galaxy Bias
 The 6th Penn State Neighborhood Workshop (2024)

Parallel Talk: Perturbation Theory Remixed
 YITP Workshop: Revisiting Cosmological Non-linearities in the Era of Precision Surveys (2023)

Presentation: Generative Artificial Intelligence and LHC Events
 KIAS QUG Summer School on A.I. in High Energy Physics (2023)

Parallel Talk: Perturbation Theory Remixed [Video Link]
 Cosmology from Home 2023 (2023)

Parallel Talk: Perturbation Theory Remixed
 The 5th Penn State Neighborhood Workshop (2023)

Parallel Talk: Exploring 21cm Intensity Mapping with Empirical Models
 The 3rd Exploring the Gas in Galaxies (EGG) Meeting at Tsinghua University (2018)

RESEARCH MENTORING

Zhaoning Liu (Ph.D. student at Tsinghua University & Visiting scholar at Penn State University, 2024-now)
Perturbative Galaxy Bias Expansion

Sophia Lieber (Undergrad student at Penn State University, 2024-now)
Machine Learning and Cosmic Structure Formation

Chengkai Yu (Undergrad student at Penn State University, 2023-now)

Lagrangian Perturbation Theory and Redshift-space Distortion

Liam Schwartz (Undergrad student at Penn State University, 2022-2023)

Cosmological Parameter Inference from Power Spectrum

Jiale Xiao (Undergrad student at Penn State University, 2022)

Julia Programming, Grid-based Calculation of Cosmological Perturbation Theory

Siyi Zhao (Undergrad student at Tsinghua University, 2020-2021)

Eulerian Perturbation Theory and Primordial Non-Gaussianity

TEACHING

Graduate Teaching Assistant at Penn State University

ASTRO 497: Data Science Applications for Exoplanets

ASTRO 415: Introduction to Astrostatistics

ASTRO 414: Stellar Structure

ASTRO 410: Computational Astrophysics

ASTRO 6: Stars, Galaxies, and the Universe (Co-teaching 20% lectures)

ASTRO 1: Astronomical Universe (Rooftop Activities and Planetarium Instructor)

SKILLS

Programming: Julia, Python, Mathematica, C, Fortran

Technologies: Git, FFT, MCMC, Machine Learning, HPC, N -body Simulations

Languages: Chinese (Native), English (Professional)

REFERENCES

Prof. Donghui Jeong (The Pennsylvania State University, US), Email: djeong@psu.edu

Prof. Atsushi Taruya (Kyoto University, Japan), Email: ataruya@yukawa.kyoto-u.ac.jp

Dr. Yun Wang (California Institute of Technology, US), Email: wang@ipac.caltech.edu